# NE723 Project 1: Discrete Ordinates Method for 1D Slab Geometry

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**** Source code available at the GitHub link below. ****

`https://github.com/zbevans3/NE723/tree/main/project-1`

# 1 Introduction

In this work, a one-speed discrete ordinates ($S_N$) neutron transport solver for steady-state, one-dimensional slab geometry with isotropic scattering and multiplying medium was developed that can be applied for fixed source and eigenvalue problems. Fixed source problems are described by Equation 1, which includes fission and an external source term.

$$\mu\frac{\partial\psi(x,\mu)}{\partial x}+\Sigma_t(x,\mu)\,\psi(x,\mu)=\frac{1}{2}\big(\Sigma_s(x)+\nu_f(x)\Sigma_f(x)\big)\int_{-1}^{1}\psi(x,\mu')d\mu'+\frac{1}{2}q(x) \tag{1}$$

where

$$x\in[0,X],\quad \mu\in[-1,1]$$

$$\psi(0,\mu)=\begin{cases}\psi_{in}^{+}(\mu) & \text{for the boundary with incoming particles}\\ \psi(0,-\mu) & \text{for the reflecting boundary}\end{cases}\quad \text{for } \mu>0$$

$$\psi(X,\mu)=\psi_{in}^{-}(\mu)\quad \text{for } \mu<0$$

$$\mu\frac{\partial\psi(x,\mu)}{\partial x}+\Sigma_t(x,\mu)\,\psi(x,\mu)=\frac{1}{2}\big(\Sigma_s(x)+\nu_f(x)\Sigma_f(x)\big)\int_{-1}^{1}\psi(x,\mu')d\mu' \tag{2}$$

where

$$x\in[0,X],\quad \mu\in[-1,1]$$

$$\psi(0,\mu)=\psi_{in}^{+}(\mu)=0\quad \text{for}\quad \mu>0$$

$$\psi(X,\mu)=\psi_{in}^{-}(\mu)=0\quad \text{for}\quad \mu<0$$

The source-free transport equation, Equation 2, describes the calculation solved in this work. The eigenvalue calculation can use reflective or vacuum boundary conditions, but only vacuum are necessary. The transport equations are discretized using Gauss–Legendre angular quadrature and the linear discontinuous Galerkin spatial method. The Source Iteration method and a power iteration are used to solve fixed source and eigenvalue problems, respectively. Three test problems examine heterogeneous media, iterative convergence, and critical size prediction.

# 2 Method

## 2.1 Discrete Ordinates Method

The Discrete Ordinates Method ($S_N$) is a determinisitc method of solving the integro-differential transport equation by dividing the angular domain into discrete angles that have a corresponding weight. The flux is then numerically integrated by multiplying an ordinate's angular flux and the quadrature weight. Applying the Discrete Ordinates Method to Equation 1, the resulting equation in 1D slab geometry in steady-state, with isotropic becomes:

$$\mu_m \frac{d\psi_m(x)}{dx} + \Sigma_t(x)\psi_m(x) = \frac{1}{2}\Big(\Sigma_s(x) + \nu_f(x)\Sigma_f(x)\Big) \sum_{m'=1}^{M} \psi_{m'}(x)\, w_{m'} + \frac{1}{2}\, q(x) \tag{3}$$

$$\psi_m(0) = \psi_{in}^{+}(\mu_m) \quad \text{for } \mu_m > 0 \quad , \quad \psi_m(X) = \psi_{in}^{-}(\mu_m) \quad \text{for } \mu_m < 0$$

where

$$\psi_m(x) = \psi(x, \mu_m),$$

$$\phi(x) = \sum_{m=1}^{M} \psi(x, \mu_m)\, w_m,$$

$$J(x) = \sum_{m=1}^{M} \mu_m\, \psi(x, \mu_m)\, w_m,$$

The choice of angular quadrature ($\mu_m$, $w_m$) is problem dependent, but in general, increasing quadrature order leads to a better approximation of the neutron flux. Equation 1 is discrete in angle but is still continuous in space. For finite computation (i.e. computers), the domain must be discretized into a finite number of cells. Several methods exist for discretizing the $S_N$ equations in space, but the method chosen in this work is the Linear discontinuous Galerkin method (LD). The LD method equations applied to Equation 3 are given as:

$$\mu_m\big(\psi_{m,i+1/2} - \psi_{m,i-1/2}\big) + \Sigma_{t,i}\, \Delta x_i\, \psi_{m,i} = \frac{\Delta x_i}{2}\Big(\big(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\big)\phi_i + q_i\Big) \tag{4}$$

$$3\big(\psi_{m,i+1/2} + \psi_{m,i-1/2} - 2\psi_{m,i}\big) + \Sigma_{t,i}\, \Delta x_i\, \widehat{\psi}_{m,i} = \frac{\Delta x_i}{2}\Big(\big(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\big)\widehat{\phi}_i + \widehat{q}_i\Big) \tag{5}$$

with the upwinding condition:

$$\psi_{m,\mp 1/2} = \begin{cases} \psi_{m,i} - \widehat{\psi}_{m,i} & , \quad \text{for } \mu_m < 0, \\ \psi_{m,i} + \widehat{\psi}_{m,i} & , \quad \text{for } \mu_m > 0, \end{cases}$$

$$i = 1, ..., N_{cells}$$

where the cell average scalar flux is:

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i}\, w_m,$$

and the linear spatial moment of the scalar flux is:

$$\widehat{\phi}_i = \sum_{m=1}^{M} \widehat{\psi}_{m,i}\, w_m,$$

Cell index $i \pm 1/2$ implies cell edge quantities, while $i$ is the cell average. Solving the angular flux in a cell or cell edge requires sweeping through the mesh, starting from known boundary condition. When stepping through the mesh, the incoming flux for cell $i$ in direction $\mu_m$ is known, therefore the four unknowns in and are reduced to three. The upwinding condition can then be used to recast the two equations into a system of two unknowns, $\psi_i$ and $\widehat{\psi}_i$.

For the positive direction ($\mu_m > 0$), the sweep begins at $x = 0$ and proceeds with $\psi_{m,i-1/2}$ being known. After substituting the upwind condition $\psi_{m,i+1/2} = \psi_{m,i} + \widehat{\psi}_{m,i}$, and become:

$$\psi_{m,i} + \hat{\psi}_{m,i} + \frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\psi_{m,i} = \frac{1}{\mu_m}\,Q_i + \psi_{m,i-1/2} \tag{6}$$

$$3\psi_{m,i} + 3\hat{\psi}_{m,i} - 6\psi_{m,i} + \frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\widehat{\psi}_{m,i} = \frac{1}{\mu_m}\,\widehat{Q}_i - 3\psi_{m,i-1/2} \tag{7}$$

where

$$Q_i = \frac{\Delta x_i}{2}\left(\left(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\right)\phi_i + q_i\right)$$

$$\widehat{Q}_i = \frac{\Delta x_i}{2}\left(\left(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\right)\widehat{\phi}_i + \widehat{q}_i\right)$$

After combining like terms, a linear system can be formed to find $\psi_{m,i}$ and $\widehat{\psi}_{m,i}$:

$$\begin{bmatrix} 1+\beta_{m,i} & 1 \\ -3 & 3+\beta_{m,i} \end{bmatrix} \begin{bmatrix} \psi_{m,i} \\ \widehat{\psi}_{m,i} \end{bmatrix} = \begin{bmatrix} \frac{1}{\mu_m}Q_i + \psi_{m,i-1/2} \\ \frac{1}{\mu_m}\widehat{Q}_i - 3\psi_{m,i-1/2} \end{bmatrix} \tag{8}$$

where

$$\beta_{m,i} = \frac{\Sigma_{t,i}\Delta x_i}{\mu_m}$$

Inverting the matrix system, leads to equations for the positive $\mu_m$ direction that need to be solved for cell $i$ to obtain the cell wise quantities. $\psi_{m,i+1/2}$ can then be computed by the unpwind condition from the newly obtained quantities.

$$\boxed{\begin{aligned} \psi_{m,i} &= \frac{(3+\beta_{m,i})Q_i - \widehat{Q}_i}{\mu_m\, G_{m,i}} + \frac{6+\beta_{m,i}}{G_{m,i}}\psi_{m,i-1/2} \\ \widehat{\psi}_{m,i} &= \frac{3Q_i + (1+\beta_{i,m})\widehat{Q}_i}{\mu_m\, G_{m,i}} - \frac{3\beta_{m,i}}{G_{m,i}}\psi_{m,i-1/2} \\ G_{m,i} &= \beta_{m,i}^2 + 4\beta_{m,i} + 6 \\ \psi_{m,i+1/2} &= \psi_{m,i} + \widehat{\psi}_{m,i} \end{aligned}} \tag{9}$$

For the negative direction ($\mu_m < 0$), the sweep begins at $x = X$ with $\psi_{m,i+1/2}$ being known. The process of finding the linear system for the positive direction is then repeated with the negative direction, instead the upwind condition is $\psi_{m,i-1/2} = \psi_{m,i} - \widehat{\psi}_{m,i}$. Upon substitution, the set of equations become:

$$-\psi_{m,i} + \hat{\psi}_{m,i} + \frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\psi_{m,i} = \frac{1}{\mu_m}\,Q_i - \psi_{m,i+1/2} \tag{10}$$

$$3\psi_{m,i} - 3\hat{\psi}_{m,i} - 6\psi_{m,i} + \frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\widehat{\psi}_{m,i} = \frac{1}{\mu_m}\,\widehat{Q}_i - 3\psi_{m,i+1/2} \tag{11}$$

which results in the linear system:

$$\begin{bmatrix} \beta_{i,m} - 1 & 1 \\ -3 & \beta_{i,m} - 3 \end{bmatrix} \begin{bmatrix} \psi_{m,i} \\ \widehat{\psi}_{m,i} \end{bmatrix} = \begin{bmatrix} \frac{1}{\mu_m} Q_i - \psi_{m,i+1/2} \\ \frac{1}{\mu_m} \widehat{Q}_i - 3\psi_{m,i-1/2} \end{bmatrix} \tag{12}$$

Thus, the equations for each of the unknowns for cell $i$ when sweeping in the negative direction ($\mu_m < 0$) are:

$$\boxed{\begin{aligned} \psi_{m,i} &= \frac{(\beta_{m,i} - 3)Q_i - \widehat{Q}_i}{\mu_m\, G_{m,i}} + \frac{6 - \beta_{m,i}}{G_{m,i}}\psi_{m,i+1/2} \\ \widehat{\psi}_{m,i} &= \frac{3Q_i + (\beta_{m,i} - 1)\widehat{Q}_i}{\mu_m\, G_{m,i}} - \frac{3\beta_{m,i}}{G_{m,i}}\psi_{m,i+1/2} \\ G_{m,i} &= \beta_{m,i}^2 - 4\beta_{m,i} + 6 \\ \psi_{m,i-1/2} &= \psi_{m,i} - \widehat{\psi}_{m,i} \end{aligned}} \tag{13}$$

## 2.2 Solution Algorithm

Herein are the algorithms required for solving a fixed source problem (Source iteration), an eigenvalue calculation (Power iteration), and a critical size calculation.

### 2.2.1 Source (inner) Iteration

From Equation 3, it is clear that the loss and source are a function of the angular flux solution. Therefore, when numerically solving the transport equation, the source must be iterated upon. The most straightforward method is simply the source iteration method (SI), in which the RHS of of Equation 3 is lagged until the flux solution converges (to within a specified tolerance). For an iterate $(l+1)$, the mesh is swept with the old scalar flux $(l)$ to obtain the $(l+1/2)$ angular flux. Once the mesh sweep is complete, the scalar the new scalar flux for $(l+1)$ is computed. The SI algorithm is provided in algorithm 1.

**Algorithm 1** Source (inner) Iteration Method

Guess initial angular and scalar flux
**while** $l \leq l_{max}$ **do**
  $l = l+1$
  Compute new source for each cell
  **for** i = 1 to $N_{cells}$ **do**
    Compute source term $Q_i^{(l)}$ and $\widehat{Q}_i^{(l)}$ from $\phi_i^{(l-1)}$, $\widehat{\phi}_i^{(l)}$
  **end for**
  Sweep in every ordinate direction
  **for** $\mu_m$ in quadrature angles **do**
    **if** $\mu_m > 0$ **then**
      Start sweep from x=0
      $\psi(0,\mu_m)^{(l)} = \psi_{in}^{+}(\mu_m)$
      **for** i = 1 to $N_{cells}$ **do**
        Solve for $\psi_{m,i}^{(l)}$, $\widehat{\psi}_{m,i}^{(l)}$, and $\psi_{m,i+1/2}^{(l)}$ using Equation 9
      **end for**
    **else if** $\mu_m < 0$ **then**
      Start sweep from x=X
      $\psi(X,\mu_m)^{(l)} = \psi_{in}^{-}(\mu_m)$
      **for** i = $N_{cells}$ to 1 **do**
        Solve for $\psi_{m,i}^{(l)}$, $\widehat{\psi}_{m,i}^{(l)}$, and $\psi_{m,i-1/2}^{(l)}$ using Equation 13
      **end for**
    **end if**
  **end for**
  Compute new scalar flux and spatial moment from $(l)$ values
  **for** $\mu_m$ in quadrature angles **do**
    **for** i = 1 to $N_{cells}$ **do**
      $\phi_i^{(l)} = \sum_{m=1}^{N_{angle}} \psi_{m,i+1/2}$
      $\widehat{\phi}_i^{(l)} = \sum_{m=1}^{N_{angle}} \widehat{\psi}_{m,i+1/2}\, w_m$
      $J_i^{(l)} = \sum_{m=1}^{N_{angle}} \mu_m\, \psi_{m,i+1/2}\, w_m$
      $\widehat{J}_i^{(l)} = \sum_{m=1}^{N_{angle}} \mu_m\, \widehat{\psi}_{m,i+1/2}\, w_m$
    **end for**
  **end for**
  Compute spectral radius of iteration: $\rho_\phi^{(l)} = \frac{||\phi^{(l)}-\phi^{(l-1)}||_\infty}{||\phi^{(l-1)}-\phi^{(l-2)}||_\infty}$
  **if** $||\phi^{(l)} - \phi^{(l-1)}||_\infty < \epsilon_\phi\left(\frac{1}{\rho_\phi^l} - 1\right)$ **then**
    Solution converged.
  **end if**
**end while**

### 2.2.2 Power (outer) Iteration

For eigenvalue problems, a power (or outer) iteration is required to determine a system's eigenvalue. In this case, an initial guess of $k_{\text{eff}}^{(0)}$ is provided to normalize the fission production (as shown in Equation 2). Then, the inner iteration solver is called to obtain a new fission source distribution. Different to a fixed source problem, the fission source term is held constant during the inner iteration and then updated after a new estimate of the eigenvalue is computed. For an outer iteration ($k$), the corresponding eigenvalue is the ratio of the new total fission source computed by the previous total fission source:

$$k_{\text{eff}}^{(k)} = k_{\text{eff}}^{(k-1)} \frac{< F\phi >^{(k)}}{< F\phi >^{(k-1)}} = k_{\text{eff}}^{(k-1)} \frac{\sum_{i=1}^{N_{cells}} \nu_{f,i} \Sigma_{f,i} \phi_i^{(k)} \Delta x_i}{\sum_{i=1}^{N_{cells}} \nu_{f,i} \Sigma_{f,i} \phi_i^{(k-1)} \Delta x_i} \tag{14}$$

The new eigenvalue is then used to modify the fission source term, and a new scalar flux is computed. This process is repeated until the eigenvalue converges to a given tolerance. The convergence in eigenvalue does not necessarily imply a fully converge flux distribution. Therefore, the convergence in fission source is also confirmed before exiting the calculation. The power iteration algorithm is provided in algorithm 2.

**Algorithm 2** Power (outer) iteration

Initialize flux vector: $\phi_i^{(0)} = 1$
Initialize $k_{\text{eff}}^{(0)} = 1$
Initialize iteration counter k = 0
Calculate initial fission source
$p_i^{(0)} = \nu_{f,i} \Sigma_{f,i} \phi_i^{(0)}$
Calculate initial total fission source
$< F\phi >^{(0)} = \sum_i^{N_{cells}} p_i^{(0)} \Delta x_i$
Start power iteration (outer)
**while** eigenvalue and fission source not converged **do**
  k = k+1
  Update old $k_{\text{eff}}$
  $k_{\text{eff}}^{(k-1)} = k_{\text{eff}}^{(k)}$
  Update old total fission source
  $< F\phi >^{(k-1)} = < F\phi >^{(k)}$
  Calculate RHS fission source term from old $k_{\text{eff}}$ and scalar flux
  $Q_{f,i}^{(k-1)} = \frac{1}{k_{\text{eff}}^{(k-1)}} p_i^{(k-1)}$
  Perform inner iteration (algorithm 1) with a fixed $Q_{f,i}$
  Calculate new fission source from $\phi$ solved from inner iteration
  $p_i^{(k)} = \nu_{f,i} \Sigma_{f,i} \phi_i^{(k)}$
  Calculate new total fission source
  $< F\phi >^{(k)} = \sum_i^{N_{cells}} p_i^{(k)} \Delta x_i$
  Calculate new eigenvalue
  $k_{\text{eff}}^{(k)} = k_{\text{eff}}^{(k-1)} \frac{< F\phi >^{(k)}}{< F\phi >^{(k-1)}}$
  Compute spectral radius of iteration: $\rho_{k\text{eff}}^{(k)} = \frac{|k_{\text{eff}}^{(k)} - k_{\text{eff}}^{(k-1)}|}{|k_{\text{eff}}^{(k-1)} - k_{\text{eff}}^{(k-2)}|}$
  **if** $|k_{\text{eff}}^{(k)} - k_{\text{eff}}^{(k-1)}| < \epsilon_k \left(\frac{1}{\rho_{k\text{eff}}^{k}} - 1\right)$ and $\max_i \frac{|p_i^{(k)} - p_i^{(k-1)}|}{p_i^{(k-1)}} < \epsilon_{pow}$ **then**
    Solution converged.
  **end if**
  Normalize scalar flux: $\sum_{i=1}^{N_{cells}} \phi_i = 1$
**end while**

## 2.3 Critical Size Calculation

For a critica size calculation, the width of the slab is iterated upon until the estimated eigenvalue reaches 1.0 (within a tolerance). An initial guess of the slab width is provied, then the eigenvalue solver is reused to obtain the system's eigenvalue. A new guess of the critical width is then computed depending on the sign of the difference in eigenvalue from critical. A simple secant search is used to efficiently guess the next critical width. The process is then repeated until the critical width is found. The critical size algorithm is provided in algorithm 3.

**Algorithm 3** Critical size algorithm

Guess initial critical size $a_{crit}^{(0)}$
Perform eigenvalue calculation to get $k^{(0)}$
$\Delta k^{(0)} = k^{(0)} - k_{crit}$
Guess new $a_{crit}$
Start critical size search
**while** Critical size not found **do**
  s = s+1
  Set slab width to $a_{crit}^{(s)}$
  Rebuild mesh with new slab width
  Perform eigenvalue calculation (algorithm 2)
  **if** $|k_{eff}^{(s)} - k_{crit}| < \epsilon_k$ **then**
    Critical size found
  **else**
    Estimate new critical size: $a_{crit}^{(s+1)} = a_{crit}^{(s)} - \Delta k^{(s)} \frac{a_{crit}^{(s)} - a_{crit}^{(s-1)}}{\Delta k^{(s)} - \Delta k^{(s-1)}}$
  **end if**
**end while**

# 3 Results

## 3.1 Test A

Test is A is a heterogeneous fixed source problem with multiplying medium. Five total material regions exist. Reflective boundary conditions are prescribed on the left side of the domain vacuum boundary on the right. A uniform spatial mesh is used with 40 total cells of width 0.20 cm. The $S_8$ Gauss-Legendre quadrature set was chosen, and a convergence criterion on the flux solution of $\epsilon_\phi = 10^6$. The material regions and cell edge locations are illustrated in Figure 1. The material cross sections and subdomain regions are provided in Table 1.

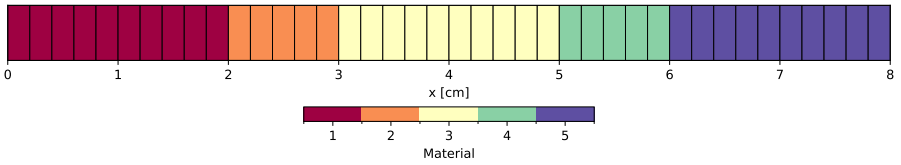



Figure 1: Test A mesh

Table 1: Test A material region definition

| Material | Right edge [cm] | $\Sigma_t$ [cm$^{-}$1] | $\Sigma_s$ [cm$^{-1}$] | $\Sigma_f$ [cm$^{-1}$] | $\nu$ [neutrons per fission] | $q$ [neutrons/cm$^3$·sec] |
|---|---|---|---|---|---|---|
| 1 | 2.0 | 50.0 | 0.0 | 0.0 | 0.0 | 50.0 |
| 2 | 3.0 | 5.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 3 | 5.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 4 | 6.0 | 1.1 | 0.7 | 0.1 | 2.0 | 0.0 |
| 5 | 8.0 | 1.1 | 0.5 | 0.16 | 2.5 | 1.0 |

After solving the system with the developed discrete ordinates transport solver, the solution converged in $\boxed{\mathrm{N}_{iter} = 42 \text{ iterations}}$. The infinity norm of the cell average scalar flux versus source iteration is shown in Figure 2. The spectral radius of convergence with respect to source iteration is provided in Figure 3. The spatial distributions of various solution variables upon convergence are given in Figure 4 through Figure 12.

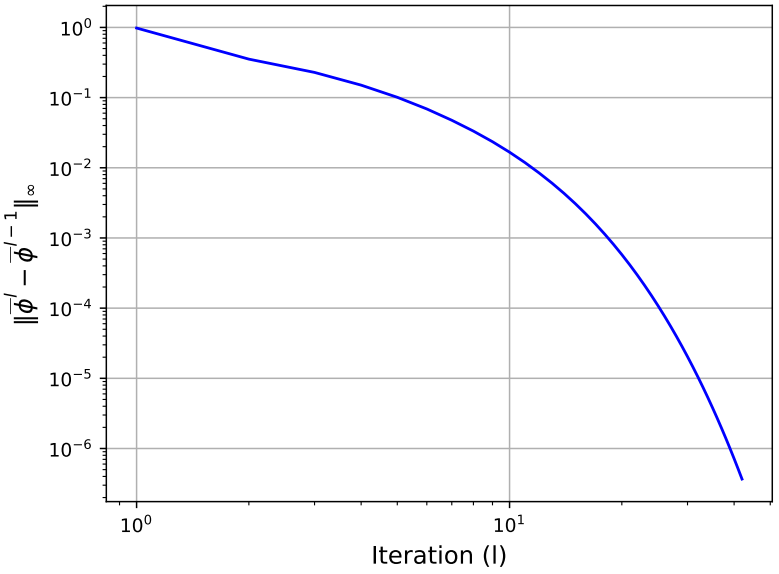



Figure 2: Test A flux error versus iterate

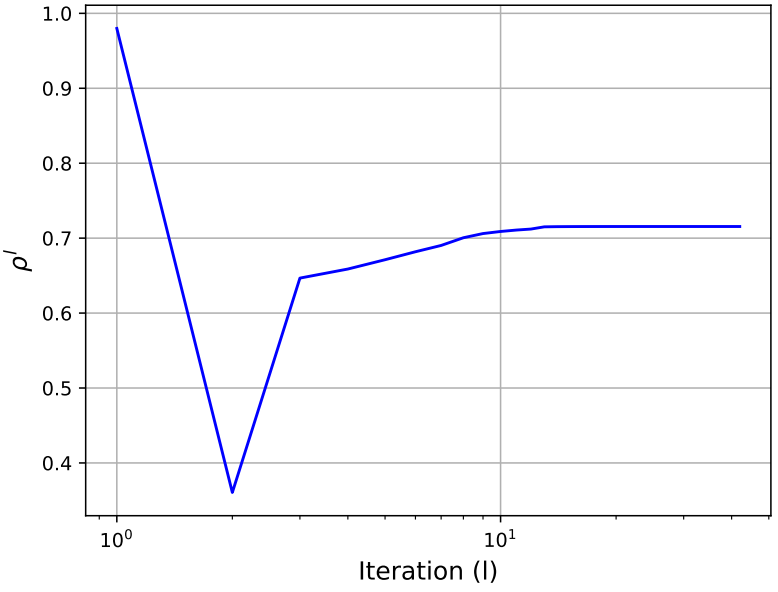



Figure 3: Test A spectral radius of iteration error versus iterate

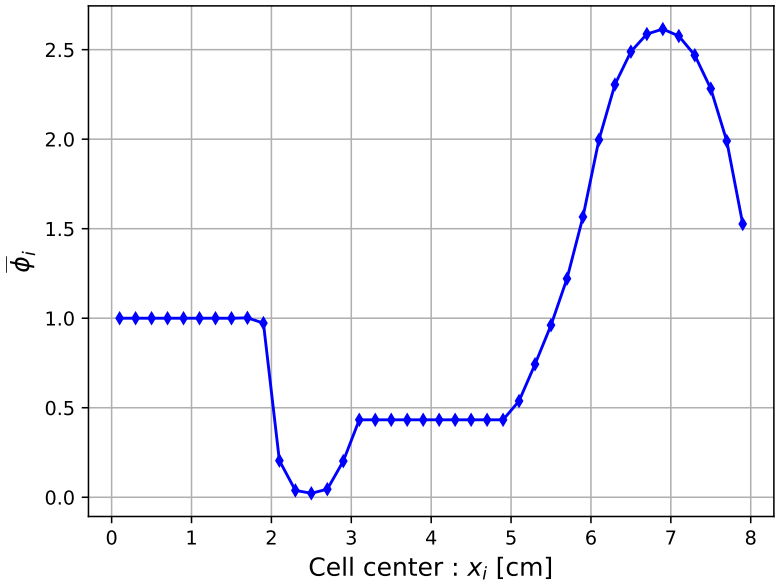



Figure 4: Test A cell average scalar flux versus position $x_i$

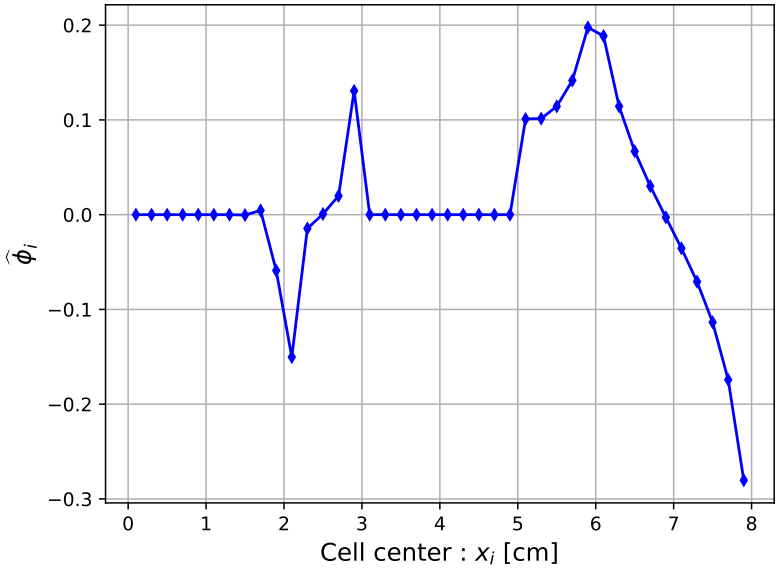



Figure 5: Test A first spatial moment of the scalar flux versus position $x_i$

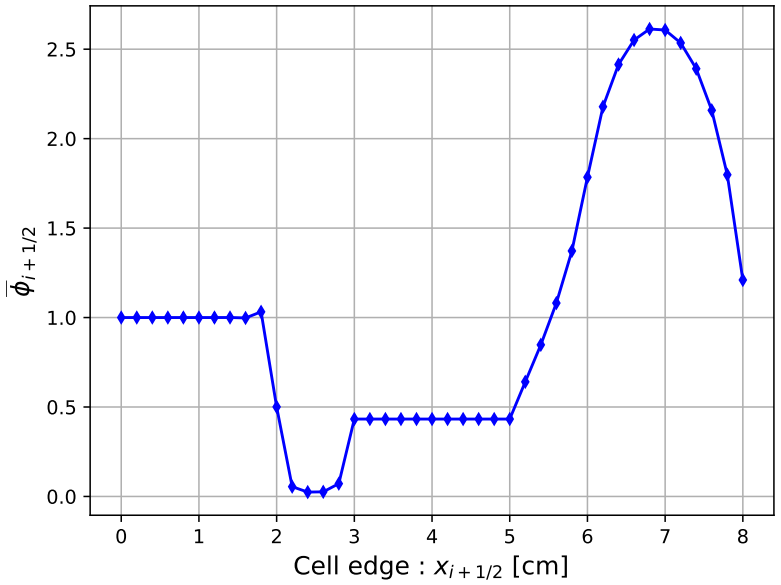



Figure 6: Test A cell edge scalar flux versus position $x_{i+1/2}$

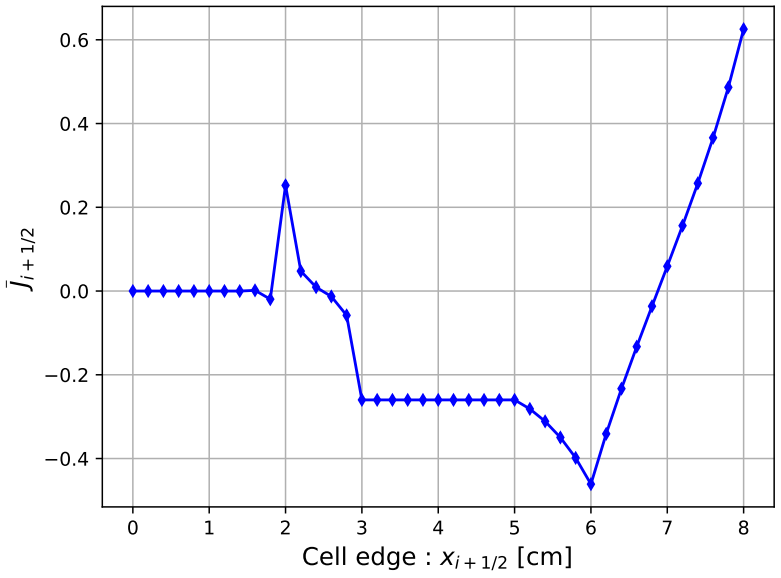



Figure 7: Test A cell edge current versus position $x_{i+1/2}$

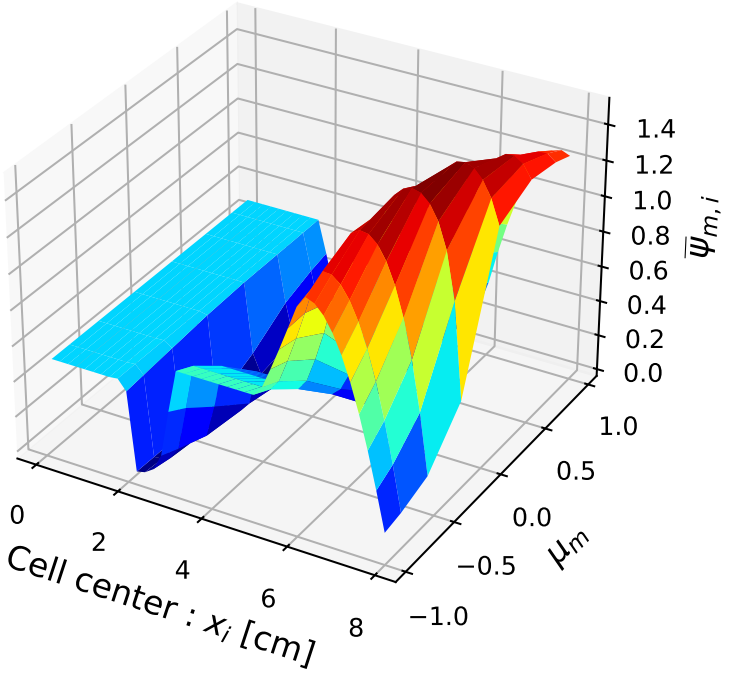



Figure 8: Test A cell average angular flux versus position and angle $(x_i, \mu_m)$

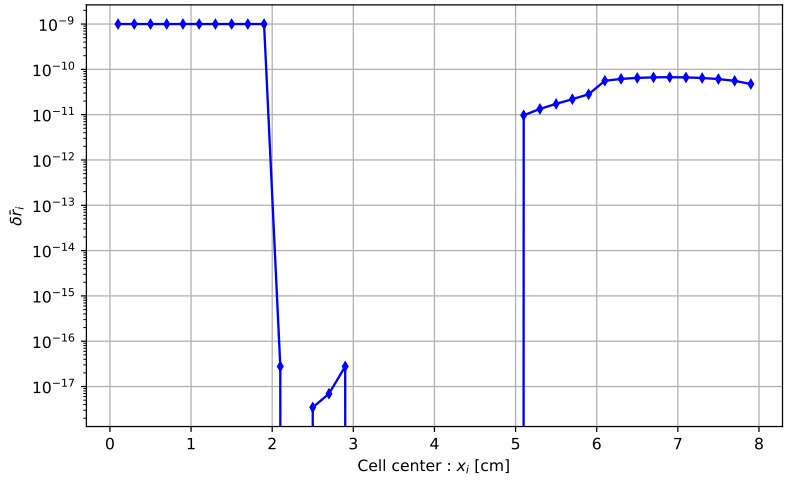



Figure 9: Test A cell average scheme residual versus position $x_i$

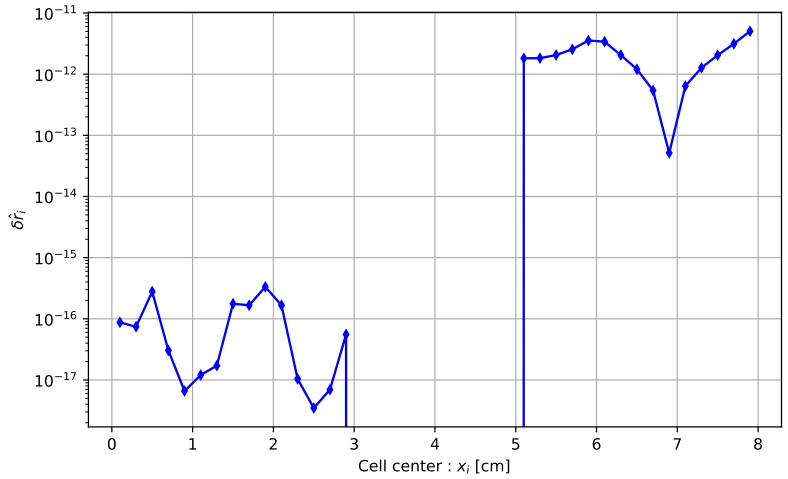



Figure 10: Test A first spatial moment scheme residual versus position $x_i$

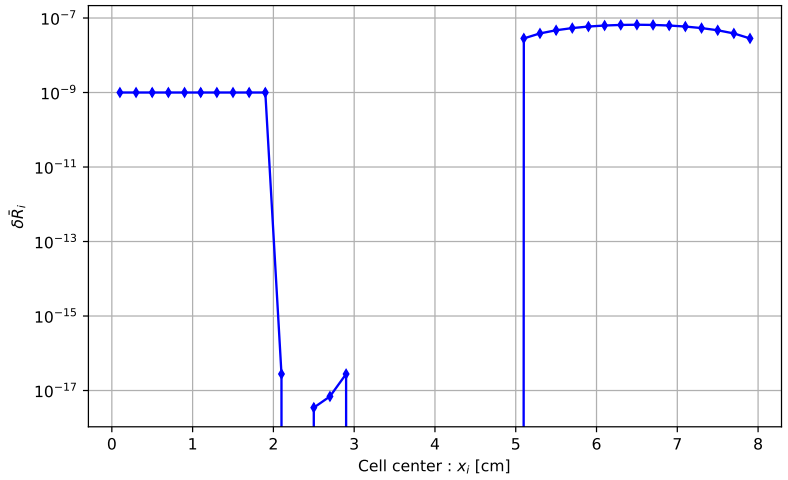



Figure 11: Test A cell average scheme iterative versus position $x_i$

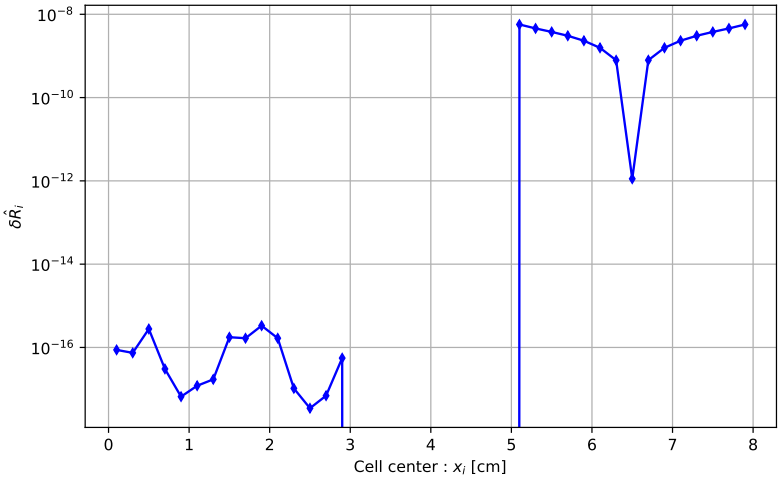



Figure 12: Test A first spatial moment iterative residual versus position $x_i$

## 3.2 Test B

Test is B is a heterogeneous fixed source problem with non-multiplying medium. Two total material regions exist. An incoming angular flux boundary conditions are prescribed on the left side with of the domain vacuum boundary on the right.

$$\psi(0, \mu) = 1.0 \quad \mu > 0$$

A non-uniform spatial mesh is used with 10 cells in the first region with width 0.10 cm and 10 cells in the second region with width 1.0 cm. The $S_8$ Gauss-Legendre quadrature set was chosen, and a convergence criterion on the flux solution of $\epsilon_\phi = 10^6$. The material regions and cell edge locations are illustrated in Figure 13. The material cross sections and subdomain regions are provided in Table 2.

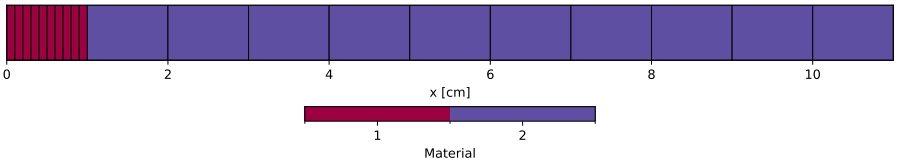



Figure 13: Test B mesh

Table 2: Test B material region definition

| Material | Right edge [cm] | $\Sigma_t$ [cm$^-$1] | $\Sigma_s$ [cm$^{-1}$] | $\Sigma_f$ [cm$^{-1}$] | $\nu$ [neutrons per fission] | $q$ [neutrons/cm$^3$·sec] |
|---|---|---|---|---|---|---|
| 1 | 1.0 | 2.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2 | 11.0 | 100.0 | 100.0 | 0.0 | 0.0 | 0.0 |

After solving the system with the developed discrete ordinates transport solver, the solution converged in $\boxed{\text{N}_{iter} = 3{,}971{,}090 \text{ iterations}}$. This is a result of the scattering ratio ($\Sigma_s/\Sigma_t$) being equal to 1.0. With very high scattering ratios, the source iteration method is known to take significant effort to converge the solution. The infinity norm of the cell average scalar flux versus source iteration is shown in Figure 14. The spectral radius of convergence with respect to source iteration is provided in Figure 15. A spectral radius close to unity implies that a solution is difficult to convergence. For Test B, the spectral radius is *very very close* to 1.0, which supports SI's slow convergence behavior. The spatial distributions of various solution variables upon convergence are given in Figure 16 through Figure 24.

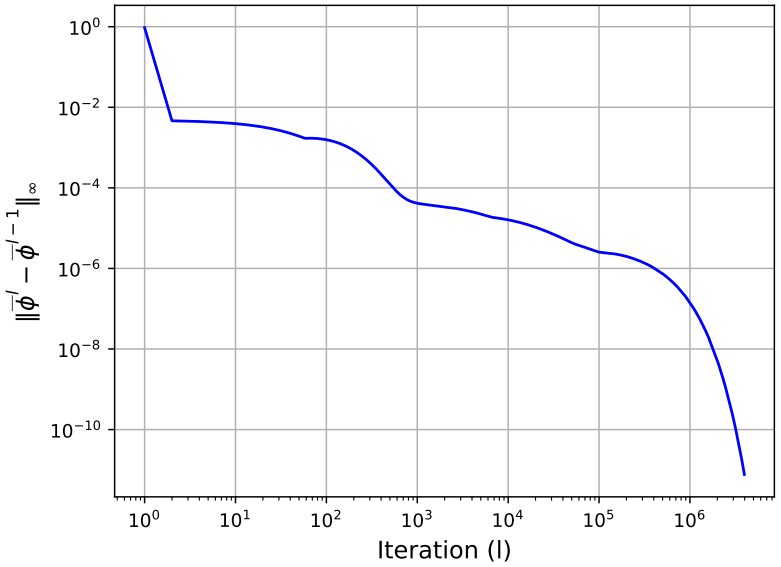



Figure 14: Test B flux error versus iterate

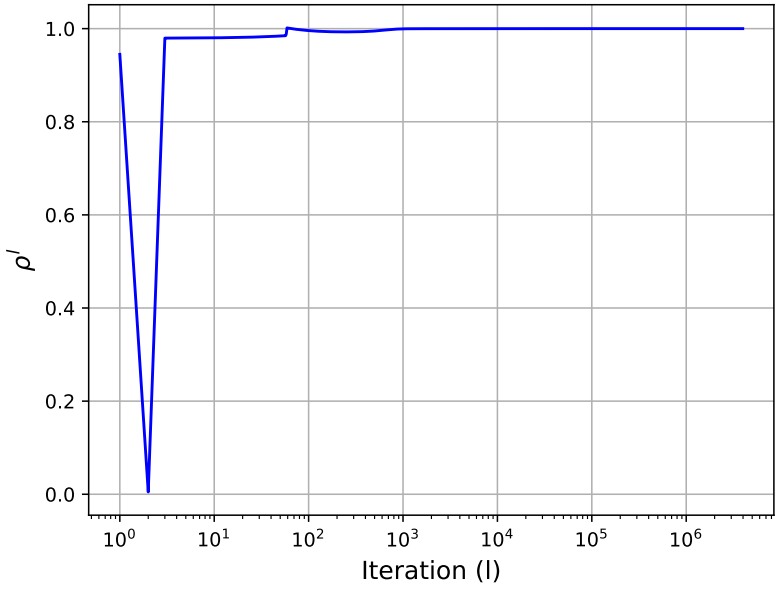



Figure 15: Test B spectral radius of iteration error versus iterate

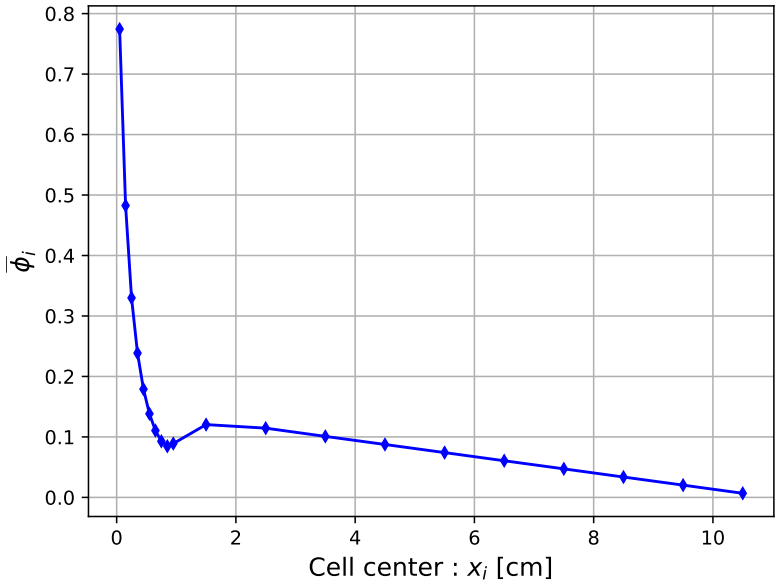



Figure 16: Test B cell average scalar flux versus position $x_i$

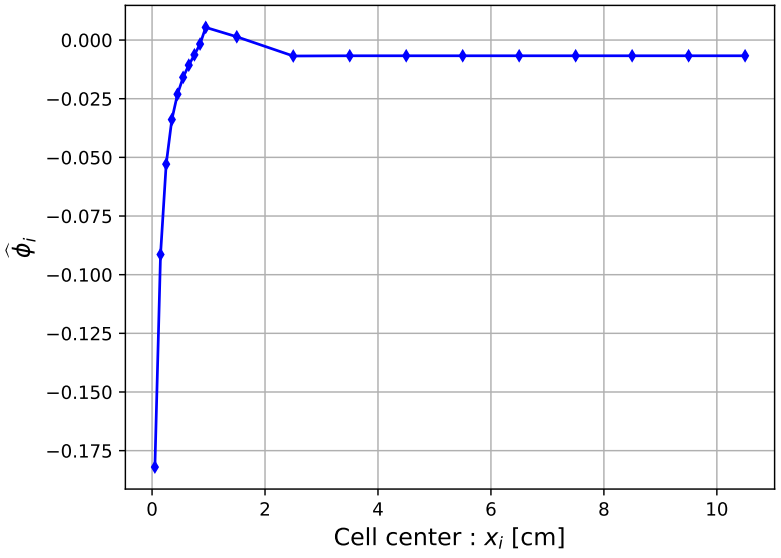



Figure 17: Test B first spatial moment of the scalar flux versus position $x_i$

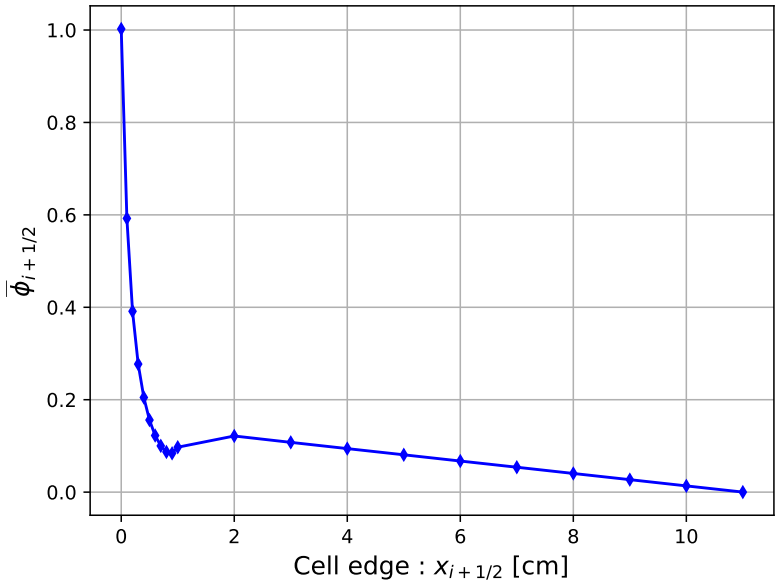



Figure 18: Test B cell edge scalar flux versus position $x_{i+1/2}$

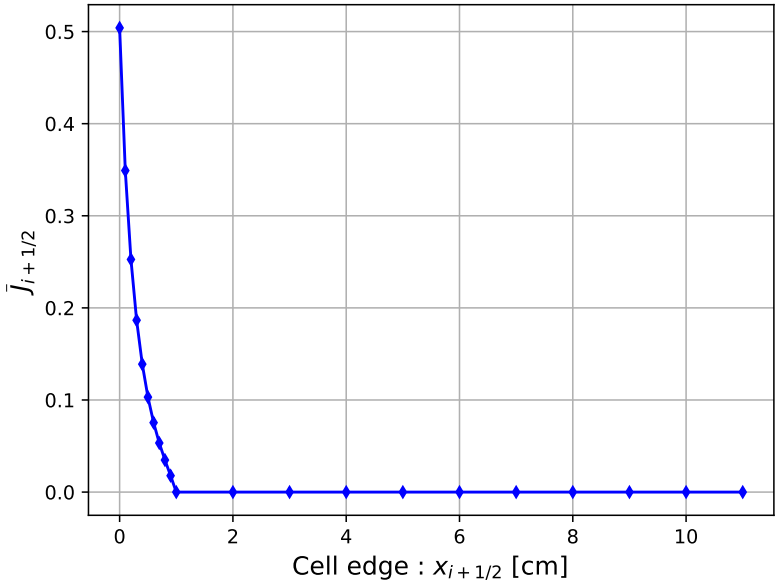



Figure 19: Test B cell edge current versus position $x_{i+1/2}$

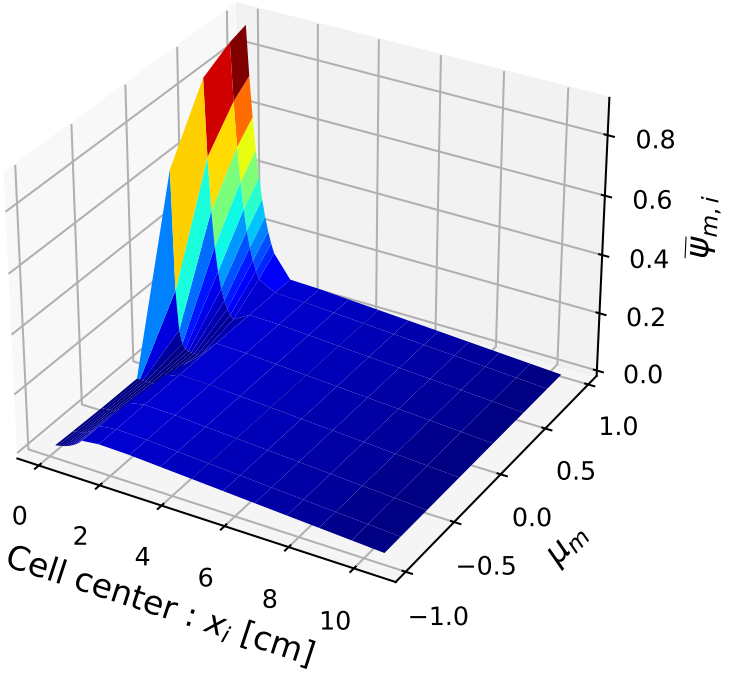



Figure 20: Test B cell average angular flux versus position and angle $(x_i, \mu_m)$

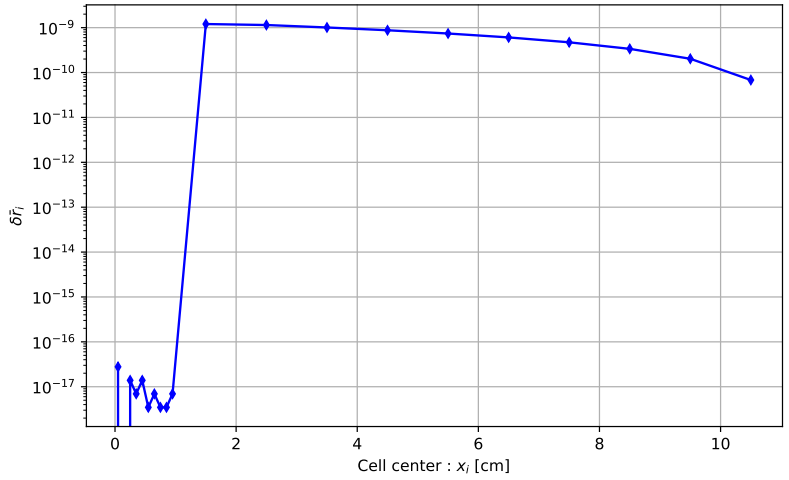



Figure 21: Test B cell average scheme residual versus position $x_i$

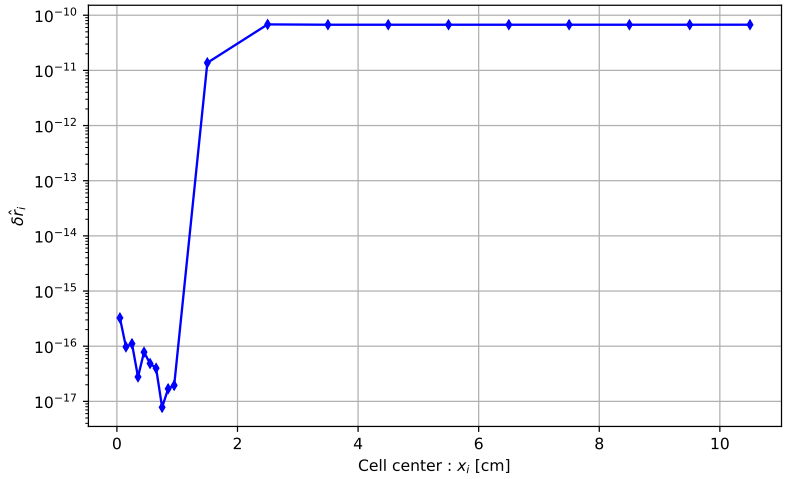



Figure 22: Test B first spatial moment scheme residual versus position $x_i$

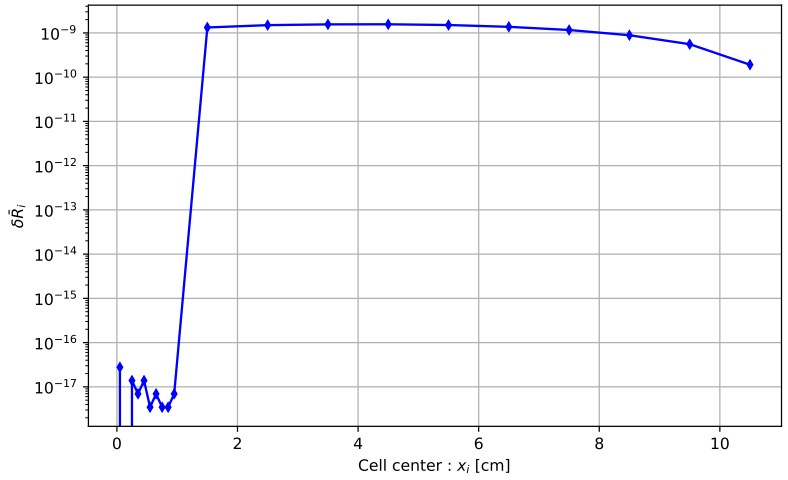



Figure 23: Test B cell average scheme iterative versus position $x_i$

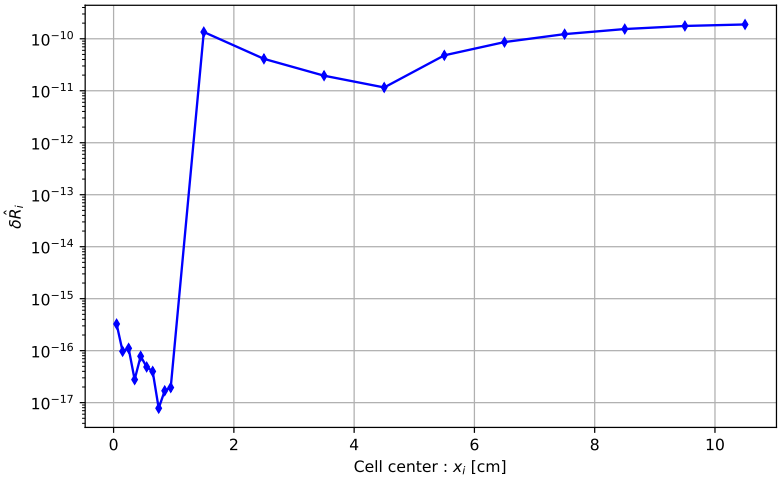



Figure 24: Test B first spatial moment iterative residual versus position $x_i$

### 3.3 Test C

Test C involves finding the critical size ($a_{crit}$) for a bare homogeneous slab. Vacuum boundary conditions were prescribed on the left and right side of the domain, and the $S_8$ Gauss-Legendre quadrature was once again used. For the calculation, the number of cells were fixed at 50, and the width of the cell was modulated until the eigenvalue converged to 1.0 (within a $10^{-6}$ tolerance between guesses). A simple secant search was used to converge on the critical size. The material properties of the slab are provided in Table 3.

Table 3: Test C material region definition

| Material | Right edge [cm] | $\Sigma_t$ [cm$^{-1}$] | $\Sigma_s$ [cm$^{-1}$] | $\Sigma_f$ [cm$^{-1}$] | $\nu$ [neutrons per fission] | $q$ [neutrons/cm$^3$·sec] |
|---|---|---|---|---|---|---|
| 1 | $a_{crit}$ | 1.0 | 0.45 | 0.3 | 2.0 | 0.0 |

The initial guess was set to a very large width of 300 cm. The critical width was found after 13 iterations with a magnitude of $\boxed{a_{crit} = 6.608123 \text{ cm}}$. The convergence in critical size is illustrated in Figure 25. The mesh corresponding to the critical size is also provided in Figure 26. We can confirm a reasonable value of $a_{crit}$ by using the one-speed eigenvalue equation from diffusion.

$$B_g^2 = \frac{\frac{\nu\Sigma_f}{k_{eff}} - \Sigma_a}{D} \tag{15}$$

where

$$\begin{aligned} k_{eff} &= 1.0 \\ D &= \frac{1}{3\Sigma_{tr}} = \frac{1}{3(\Sigma_t - \bar{\mu}\Sigma_{s1})} = \frac{1}{3\Sigma_t} \\ \Sigma_a &= \Sigma_t - \Sigma_s \end{aligned}$$

Since isotropic scattering is assumed, the transport cross section ($\Sigma_{tr}$) is equivalent to the total cross section ($\Sigma_t$). For 1D slab geometry, $B_g^2 = \left(\frac{\pi}{a}\right)^2$, so $a_{crit}^{diffusion}$ can be computed. Since the diffusion approximation uses an extrapolated boundary, the computed critical size from the geometric buckling can be transformed to an approximate transport solution by:

$$a_{crit} = a_{crit}^{diffusion} - 2 \cdot 0.7104 \cdot \left(\frac{1}{\Sigma_{tr}}\right) \tag{16}$$

The 0.7104 comes from a transport derived extrapolation distance for plane geometries. Combining these equations, an estimated critical size by the one-speed equation is 6.6908 cm. The result computed by the discrete ordinates solver closely matches, and the slight difference could be down to the one-speed approximation and discretization error. The spatial distributions of various solution variables upon convergence are given in Figure 27 through Figure 35.

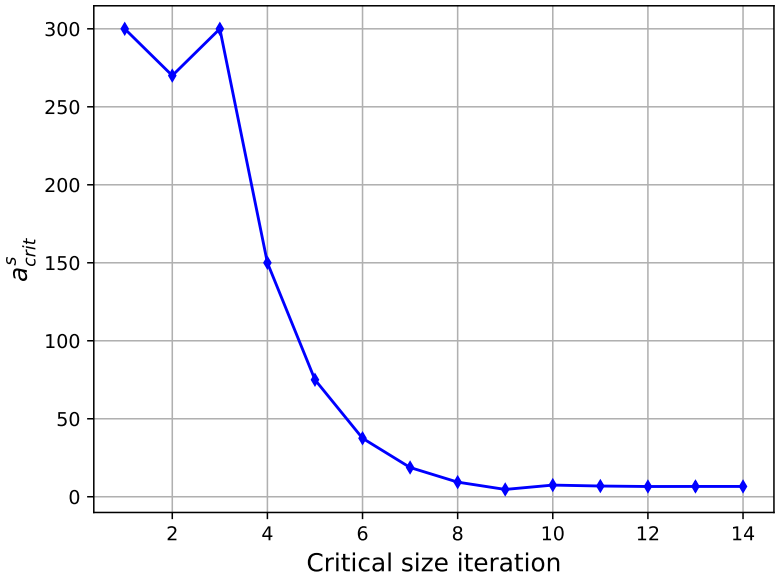



Figure 25: Test C critical size versus iterate

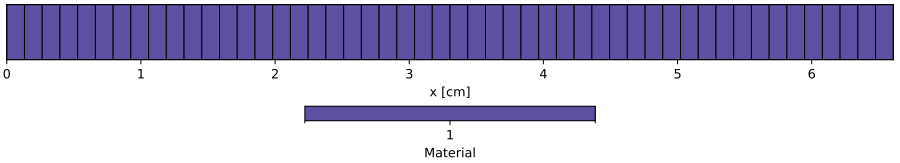



Figure 26: Test C mesh

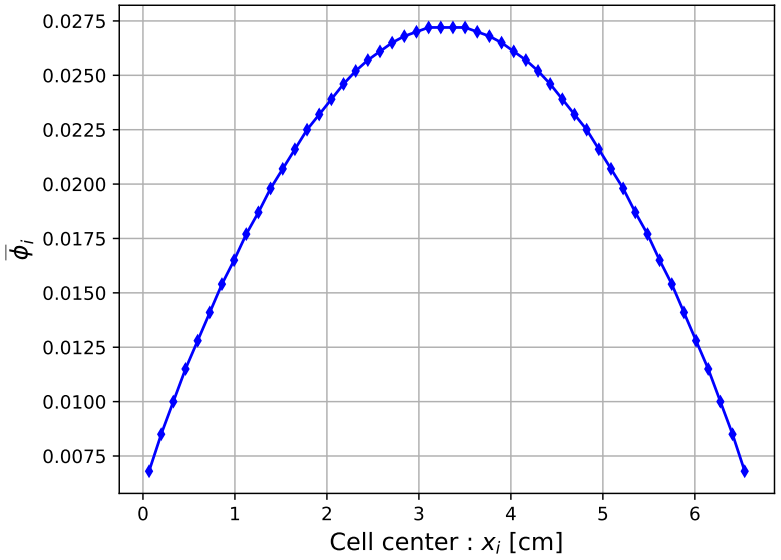



Figure 27: Test C cell average scalar flux versus position $x_i$

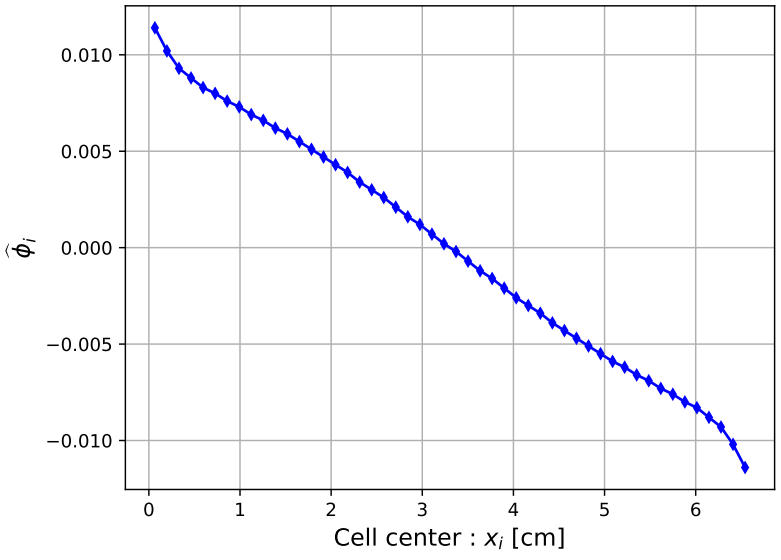



Figure 28: Test C first spatial moment of the scalar flux versus position $x_i$

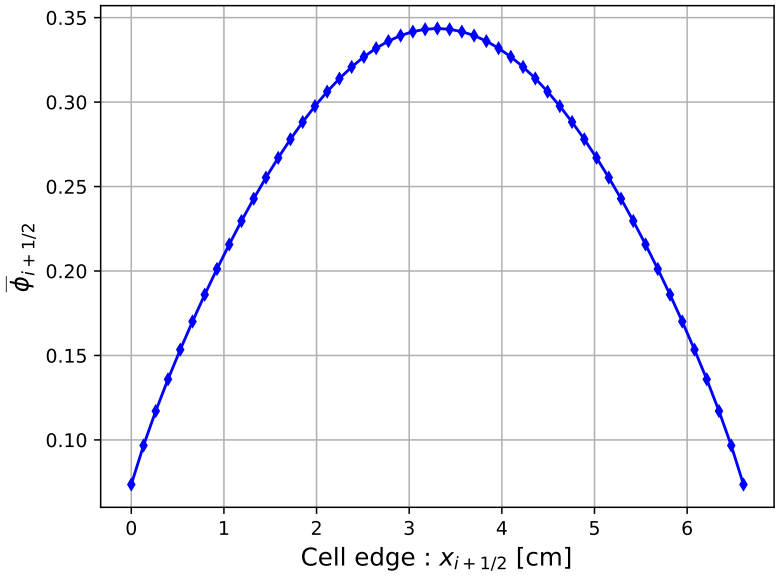



Figure 29: Test C cell edge scalar flux versus position $x_{i+1/2}$

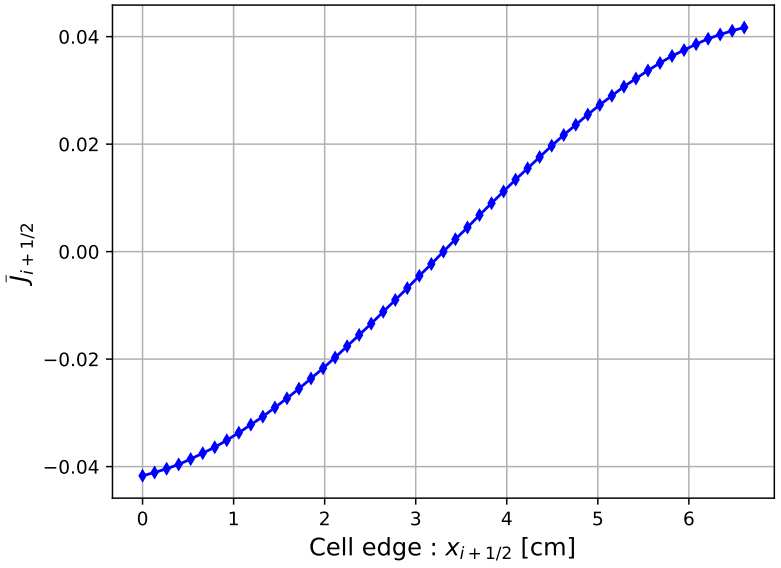



Figure 30: Test C cell edge current versus position $x_{i+1/2}$

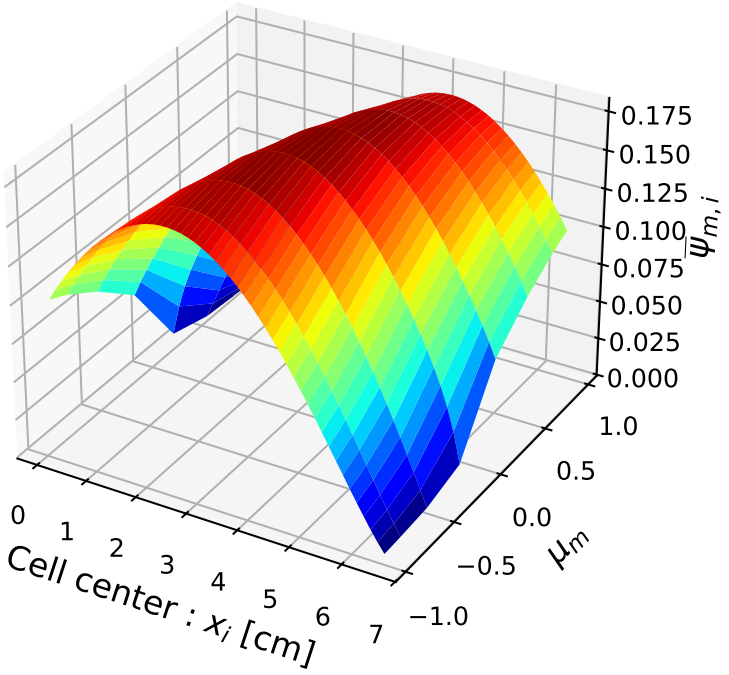



Figure 31: Test C cell average angular flux versus position and angle $(x_i, \mu_m)$

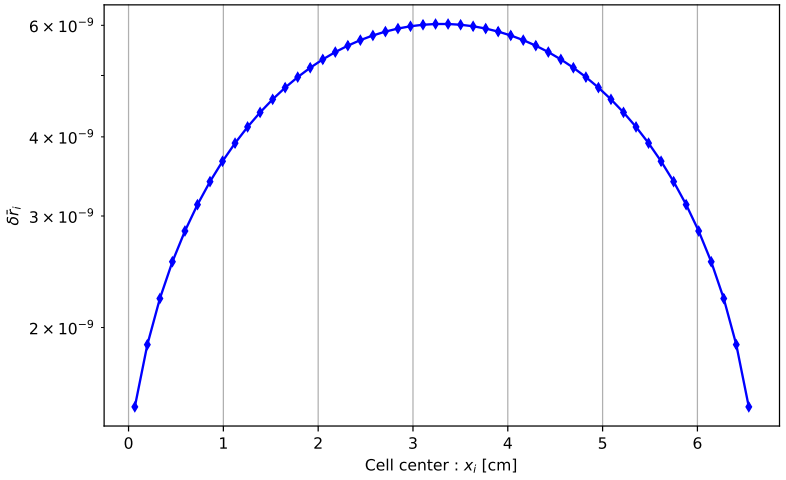



Figure 32: Test C cell average scheme residual versus position $x_i$

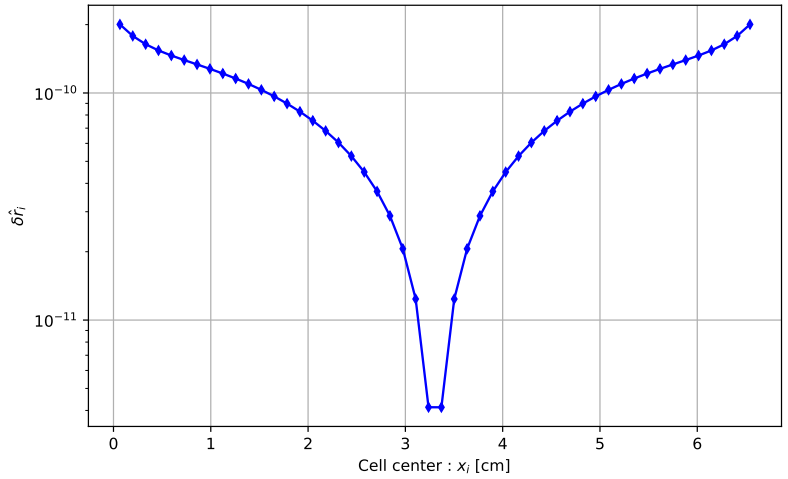



Figure 33: Test C first spatial moment scheme residual versus position $x_i$

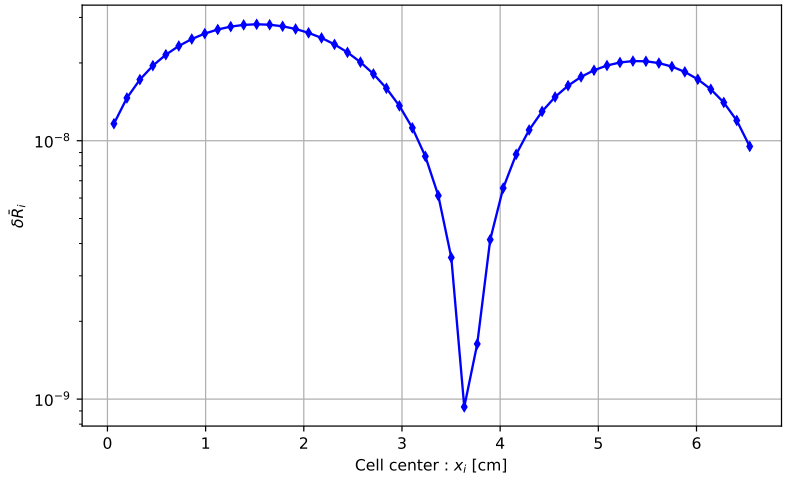



Figure 34: Test C cell average scheme iterative versus position $x_i$

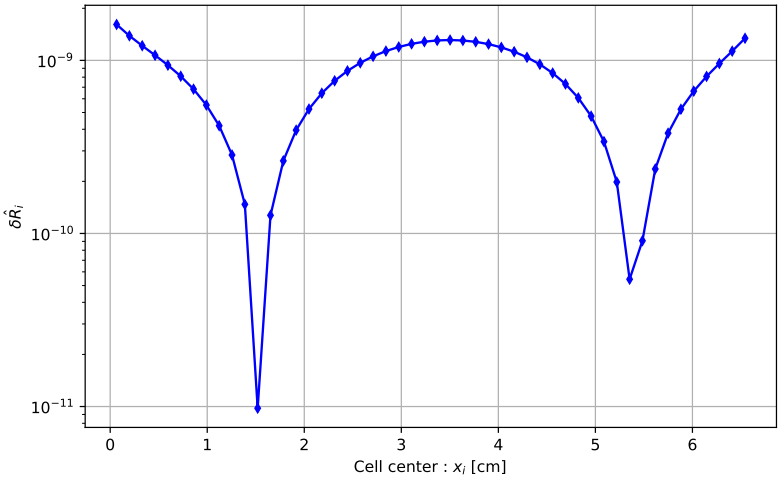



Figure 35: Test C first spatial moment iterative residual versus position $x_i$

# 4 Summary

A one-dimensional $S_N$ transport solver was developed using linear discontinuous spatial discretization and applied to three test problems. The fixed source calculations demonstrated the treatment of heterogeneous materials and different boundary conditions. Test B highlighted the slow convergence of unaccelerated source iteration in a purely scattering region, requiring approximately 3.97 million iterations as the spectral radius approached unity. For Test C, the secant search obtained a critical slab width of 6.608123 cm, compared with the diffusion-based estimate of 6.6908 cm, a difference of approximately 1.24%. The reported residuals provide checks on the discrete equations and iterative convergence, while the critical size comparison supports the reasonableness of the transport result.